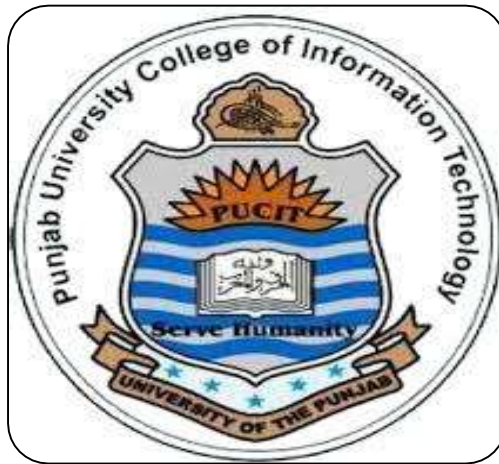


PROJECT PROPOSAL

Information Security Project Proposal



TOPIC:

Multi-Layer Canary Token Based Intrusion Detection and Alert System

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1. Problem Statement / Project Idea

- Attackers or malicious insiders often try to access sensitive files like passwords, system keys, and configuration files.
- Traditional security measures may prevent unauthorized login but cannot detect when someone is exploring the system for sensitive data.
- This project uses canary tokens (decoy files) to detect suspicious activity.
- When a decoy file is accessed, modified, or deleted, the system:
 - Generates a real-time alert
 - Logs the activity with file name, action, and timestamp
- Objective: Demonstrate how deception-based security can detect and prevent unauthorized access inside a system.

2. Scope

The project simulates a decoy-based intrusion detection system using canary files. It monitors unauthorized file access, triggers alerts, and logs suspicious activity. Designed for a controlled environment using simple C++ file handling and logging.

Project Coverage:

- Automatic generation of decoy (canary) files.
- Monitoring of file access, modification, and deletion.
- Real-time alerts on suspicious activity.
- Logging of events including file name, action, and timestamp.

Limitations / Boundaries:

- Simulation is limited to the decoy files created within the system.
- Does not include machine learning or advanced threat detection.
- Focused on controlled environment; not designed for enterprise-level network monitoring.
- Detection depends on rule-based triggers (file access, modify, delete) and may not catch sophisticated attacks outside these rules.